# Representation of Data

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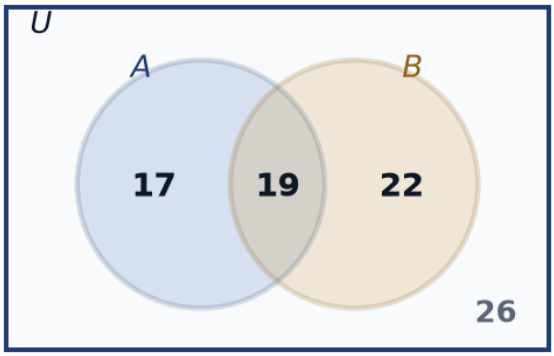
## Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

**1** [Foundational] • Target: Stem-and-leaf diagrams

The data set $23, 25, 31, 34, 34, 38, 42, 45$ is to be shown as a stem-and-leaf diagram with stems $2, 3, 4$. Write out the diagram, and use it to find the median.

**2** [Foundational] • Target: Stem-and-leaf diagrams

The data set $17, 19, 22, 26, 26, 29, 33, 37, 41$ is to be shown as a stem-and-leaf diagram with stems $1, 2, 3, 4$. Write out the diagram, and use it to find the range.

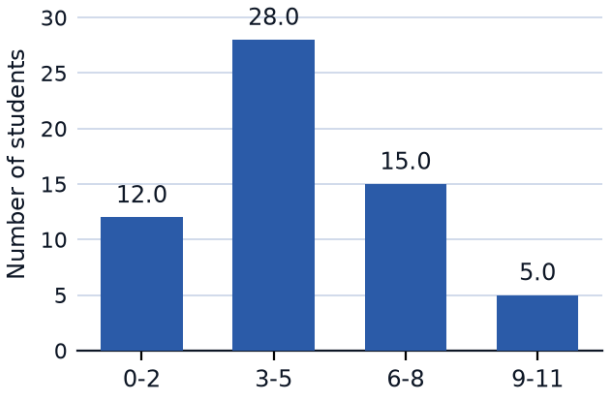



**3** [Foundational] • Target: Histograms and frequency density

A class interval of width 10 contains 25 items. Find the frequency density, which is plotted on the vertical axis of a histogram.

**4** [Foundational] • Target: Reading grouped frequency data from a bar chart

The bar chart shows the number of books read by students in a reading challenge, grouped by number of books. Find the total number of students, and the modal group.



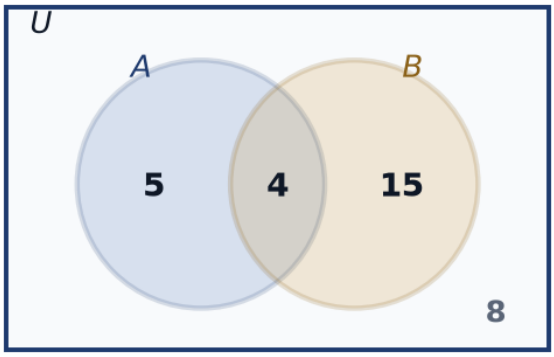

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**5** [Foundational] • Target: Reading grouped frequency data from a bar chart

Using the same reading-challenge data (0-2: 12, 3-5: 28, 6-8: 15, 9-11: 5 students), find what percent of students read 6 or more books.

**6** [Foundational] • Target: Measures of central tendency for grouped data

A grouped data set has class midpoints and frequencies: $5$ (freq $4$), $15$ (freq $8$), $25$ (freq $6$), $35$ (freq $2$). Estimate the mean.




**7** [Foundational] • Target: Box-and-whisker plots

A data set has minimum $= 8$, $Q_1 = 15$, median $= 22$, $Q_3 = 30$, and maximum $= 40$. State the interquartile range, and determine whether a value of $45$ would be considered an outlier if outliers are defined as values beyond $Q_3 + 1.5 \times IQR$.

## Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

**8** [Intermediate] • Target: Box-and-whisker plots

A data set has minimum $= 5$, $Q_1 = 12$, median $= 18$, $Q_3 = 27$, and maximum $= 50$. State the interquartile range, and determine whether the maximum value would be considered an outlier if outliers are defined as values beyond $Q_3 + 1.5 \times IQR$.

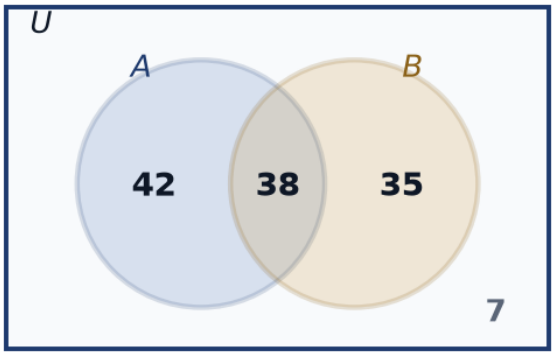
**9** [Intermediate] • Target: Box-and-whisker plots

A data set has $Q_1 = 20$ and $Q_3 = 32$. Find the lower outlier boundary, defined as any value below $Q_1 - 1.5 \times IQR$.

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**10** [Intermediate] • Target: Skewness from summary statistics

A data set has mean $= 42$, median $= 38$, and mode $= 35$. Describe the skewness of the distribution.




**11** [Intermediate] • Target: Comparing distributions using summary measures

Two classes take the same test. Class A has mean $68$ and standard deviation $5$. Class B has mean $71$ and standard deviation $12$. Compare the two classes' performance and consistency.

**12** [Intermediate] • Target: Comparing distributions using summary measures

Two machines fill bottles with a target volume of 500 mL. Machine X has mean fill volume $500$ mL and standard deviation $2$ mL. Machine Y has mean fill volume $498$ mL and standard deviation $8$ mL. Compare the two machines' accuracy and consistency.

**13** [Intermediate] • Target: A well-designed word problem: analyzing delivery times

This question has two parts. A delivery company records the delivery times (in minutes) for 50 packages: mean $= 34$ minutes, standard deviation $= 6$ minutes. Company policy states any delivery beyond 2 standard deviations above the mean is classified as late. (a) Find the delivery-time threshold, in minutes, above which a delivery is classified as late. (b) If a customer's package took 44 minutes to deliver, determine, using the standard deviation, whether this delivery should be classified as late, and state how many standard deviations from the mean this value lies.

**14** [Intermediate] • Target: Skewness from summary statistics

Find all complex solutions to $z^4 + 16 = 0$ in exact rectangular form $a + bi$ using De Moivre's roots theorem.

## Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

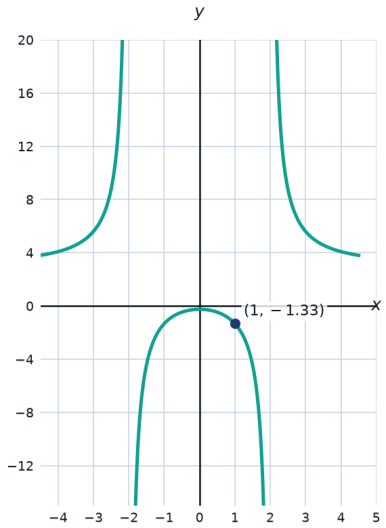
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**15** [Challenging] • Target: Comparing distributions using summary measures • Cross-Lesson: 11.01 Kinematics

The depth of water $h(t)$ in meters at a harbor entrance $t$ hours after midnight is modelled by $h(t) = 3.5\cos\left(\frac{\pi t}{6}\right) + 6.5$ for $0 \le t \le 24$. (a) State the maximum and minimum depths of water and their times of occurrence. (b) Find the period of the tidal oscillation. (c) A cargo ship requires a minimum water depth of 8.25 meters to safely enter the harbor. Find the total number of hours during the first 12 hours of the day when the ship can safely enter.

**16** [Challenging] • Target: A well-designed word problem: analyzing delivery times • Cross-Lesson: 12.01 Forces & Equilibrium

Find all vertical, horizontal, and slant asymptotes and any removable discontinuities (holes) of the rational function $f(x) = \frac{3x^2 + 1}{x^2 - 4}$. Describe its end behavior using limits.




**17** [Challenging] • Target: Stem-and-leaf diagrams • Cross-Lesson: 13.01 Newton's Laws

A circle with centre $O$ has radius $r = 8$ cm. Chord $AB$ subtends a central angle $\angle AOB = \frac{3\pi}{4}$ radians. (a) Find the exact length of the minor arc $AB$. (b) Find the exact area of the sector $OAB$. (c) Find the exact area of the minor segment enclosed between chord $AB$ and arc $AB$.

**18** [Challenging] • Target: Histograms and frequency density • Cross-Lesson: 14.01 Energy, Work & Power

Decompose the rational expression $R(x) = \frac{7x - 1}{x^2 - x - 6}$ into partial fractions using the Heaviside cover-up method.

**19** [Challenging] • Target: Reading grouped frequency data from a bar chart • Cross-Lesson: 16.01 Permutations & Combinations

Evaluate the definite integral $\int_0^{\sqrt{7}} \frac{4x}{\sqrt{x^2 + 9}}\,dx$ using the substitution $u = x^2 + 9$.

**20** [Challenging] • Target: Measures of central tendency for grouped data • Cross-Lesson: 17.01 Discrete Random Variables

A surveyor measures distances from an observation point $C$ to two landmarks $A$ and $B$: $AC = 140$ m and $BC = 210$ m, with included angle $\angle ACB = 78^\circ$. Find the distance $AB$ between the landmarks to the nearest tenth of a meter, and compute the area of $\Delta ABC$ to the nearest square meter.

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## Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

**21** [Foundational] • Target: Stem-and-leaf diagrams

A grouped frequency table has class intervals of width $10$ starting at $12$, with frequencies $[10, 4, 5, 6]$ respectively. Using midpoints, find the mean.

**a.** 149/5

**b.** 144/5

**c.** 154/5

**d.** 159/5

**22** [Foundational] • Target: Histograms and frequency density

In a histogram, the frequency density for a class of width 4 with frequency 20 is:

**a.** 4

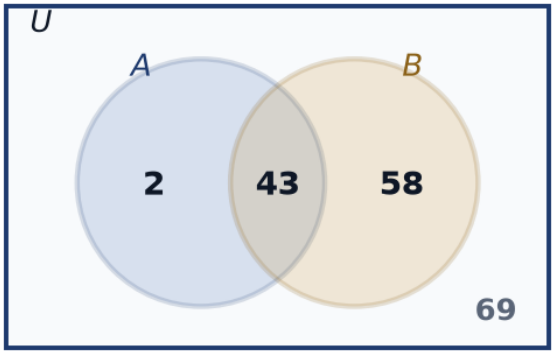
**b.** 5

**c.** 8

**d.** 10

**23** [Foundational] • Target: Reading grouped frequency data from a bar chart

For the data set $2, 43, 58, 69, 87, 92, 96$, find the upper boundary for outliers, using $Q_1 - 1.5\,IQR$ and $Q_3 + 1.5\,IQR$.




**a.** 325/2

**b.** 331/2

**c.** 345/2

**d.** 337/2

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**24** [Foundational] • Target: Measures of central tendency for grouped data

For the data set $11, 45, 57, 65, 69, 71, 91$, find the lower boundary for outliers, using $Q_1 - 1.5\,IQR$ and $Q_3 + 1.5\,IQR$.

**a.** 6

**b.** 9

**c.** 3

**d.** 13

**25** [Foundational] • Target: Box-and-whisker plots

Find the median of the data set: $8, 25, 29, 42, 47, 49, 54$.

**a.** 46

**b.** 37

**c.** 42

**d.** 41

**26** [Foundational] • Target: Skewness from summary statistics

Find the range of the data set: $7, 7, 17, 20, 18, 22, 17$.

**a.** 13

**b.** 20

**c.** 15

**d.** 17

**27** [Foundational] • Target: Comparing distributions using summary measures

Find the mean of the data set: $7, 11, 3, 2$.

**a.** 23/4

**b.** 19/4

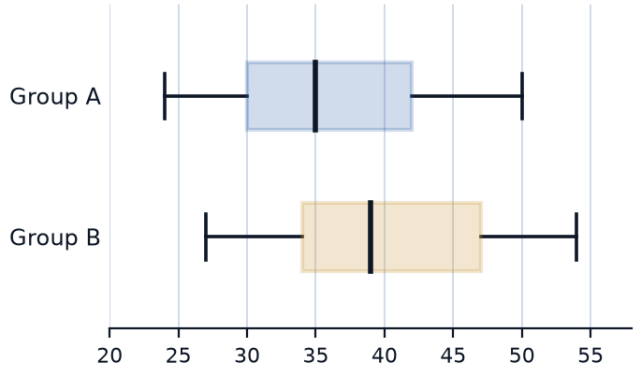
**c.** 27/4

**d.** -23/4

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**28** [Foundational] • Target: A well-designed word problem: analyzing delivery times

Find the interquartile range of the data set: $24, 29, 32, 34, 57, 74, 91$.




**a.** 45

**b.** 47

**c.** 44

**d.** 50

## Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

**29** [Intermediate] • Target: Stem-and-leaf diagrams

The following bar chart shows the number of students who achieved different grades in a mathematics test. The grades are A, B, C, and D. The data is as follows: Grade A: 14 students, Grade B: 22 students, Grade C: 18 students, Grade D: 26 students. What is the total number of students who took the test?

**a.** 70

**b.** 80

**c.** 78

**d.** 60

**30** [Intermediate] • Target: Histograms and frequency density

In a class of $20$ students, $8$ study only French, $7$ study only Spanish, $3$ study both, and $2$ study neither. A student is picked at random. Find the probability that the student studies French.

**a.** 1/2

**b.** 4/5

**c.** 11/20

**d.** 7/8

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**31** [Intermediate] • Target: Reading grouped frequency data from a bar chart

A bag contains $15$ balls, of which $7$ are red. Two balls are drawn at random, one after another, without replacement. Find the probability that both are red.

**a.** 15/16

**b.** 17/21

**c.** 1/5

**d.** 3/4

**32** [Intermediate] • Target: Measures of central tendency for grouped data

A bag contains $8$ balls, of which $3$ are red. Two balls are drawn at random, one after another, with replacement. Find the probability that both are red.

**a.** 9/64

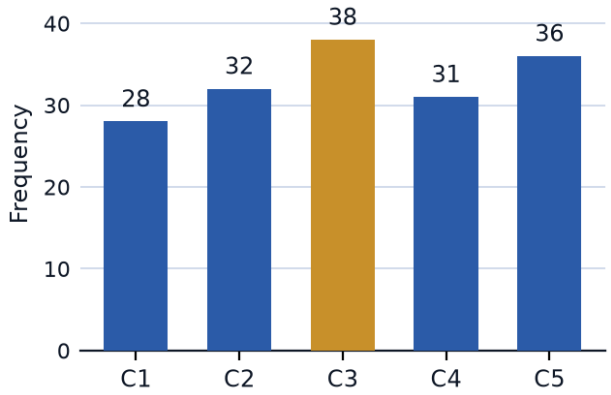
**b.** 13/23

**c.** 1/2

**d.** 7/12

**33** [Intermediate] • Target: Box-and-whisker plots

Events $A$ and $B$ are such that $P(A) = 3/10$, $P(B) = 3/10$, and $P(A \cap B) = 1/10$. Find $P(A \cup B)$.




**a.** 3/7

**b.** 4/17

**c.** 17/21

**d.** 1/2

**34** [Intermediate] • Target: Skewness from summary statistics

A bag contains $11$ counters, of which $3$ are red. A counter is drawn at random. Find the probability that it is red.

**a.** 2/3

**b.** 3/4

**c.** 3/11

**d.** 11/15

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**35** [Intermediate] • Target: Comparing distributions using summary measures

The box plot below shows the distribution of scores for a class on a recent statistics quiz. The five-number summary is given by minimum, first quartile (Q1), median, third quartile (Q3), and maximum. If the interquartile range (IQR) is 25, and the median score is 70, find the value of Q1.

**a.** 45 **b.** 60

**c.** 75 **d.** 95

**36** [Intermediate] • Target: A well-designed word problem: analyzing delivery times

The box plot below shows the distribution of scores for a class of 30 students on a recent test. The interquartile range (IQR) is 22 marks. The median score is 75 marks. The minimum score is 30 marks, and the maximum score is 98 marks. Let the first quartile be denoted by $Q_1$ and the third quartile be denoted by $Q_3$. If the upper whisker extends to 98 marks, and the third quartile $Q_3$ is 86 marks, find the value of $Q_1$ .

**a.** 64 **b.** 75

**c.** 80 **d.** 53

**37** [Intermediate] • Target: Stem-and-leaf diagrams

The bar chart below shows the number of students who chose each of four different sports as their favorite. What is the total number of students surveyed?

**a.** 50 **b.** 60

**c.** 70 **d.** 80

## Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

**38** [Challenging] • Target: Histograms and frequency density • Cross-Lesson: 14.01 Energy, Work & Power

The bar chart below shows the number of students who achieved a certain score on a test. What is the total number of students who scored 6 or 8?

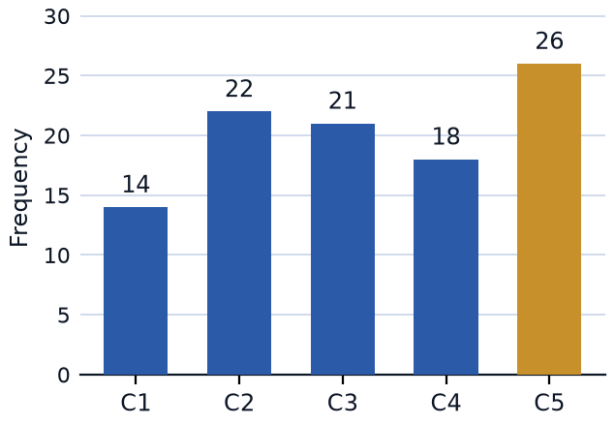
**a.** 30 **b.** 28

**c.** 26 **d.** 32

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**39** [Challenging] • Target: Reading grouped frequency data from a bar chart • Cross-Lesson: 16.01 Permutations & Combinations

The point $(3, -2)$ lies on the curve $y = f(x)$. Find the coordinates of the corresponding point on $y = f(x + 5) - 2$.




**a.** (-2,-4)

**b.** (-2,4)

**c.** (3,-2)

**d.** (-4,-2)

**40** [Challenging] • Target: Measures of central tendency for grouped data • Cross-Lesson: 17.01 Discrete Random Variables

A discrete random variable $X$ has probability distribution given by $P(X = 1) = k, P(X = 2) = k, P(X = 3) = 2k$, where $k$ is a constant. Find $k$.

**a.** 1/3

**b.** 1/4

**c.** 1/5

**d.** 1/2

**41** [Challenging] • Target: Box-and-whisker plots • Cross-Lesson: 18.01 Binomial Distribution

Solve the simultaneous equations $2x + 3y = 18$ and $3x - 4y = 10$.

**a.** x=7, y=2

**b.** x=-6, y=-2

**c.** x=6, y=2

**d.** x=2, y=6

**42** [Challenging] • Target: Skewness from summary statistics • Cross-Lesson: 19.01 Normal Distribution

A line has gradient $-2$. Find the gradient of a line perpendicular to it.

**a.** 2

**b.** -1/2

**c.** -2

**d.** 1/2

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**43** [Challenging] • Target: Comparing distributions using summary measures • Cross-Lesson: 2.01 Simultaneous Equations & Inequalities

Find the area enclosed between the curve $y = x$, the $x$-axis, and the lines $x = 0$ and $x = 3$.

**a.** 19/2

**b.** 9

**c.** 9/2

**d.** 7/2

**44** [Challenging] • Target: A well-designed word problem: analyzing delivery times • Cross-Lesson: 3.01 Graphs & Transformations

Find the equation of the line through $(-1, -4)$ with gradient $-2$, in the form $y = mx + c$.

**a.** y=2x-6

**b.** y=-2x-5

**c.** y=-2x-6

**d.** y=-2x+6

**45** [Challenging] • Target: Stem-and-leaf diagrams • Cross-Lesson: 4.01 Coordinate Geometry

The function $f(x) = \frac{1}{x-4}$ is undefined at one value of $x$. Find this value.

**a.** x=0

**b.** x=-4

**c.** x=5

**d.** x=4

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